Network-based Multi-omics Integration Workflow Linking Transcriptomics and Phenotypic Traits in Hanwoo cattle under methane-mitigating feeding strategies

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Background

Omics technologies, such as RNA-seq, generate extensive molecular information; yet, many analyses stop at listing differentially expressed genes or enriched pathways, without explaining how these molecular changes are connected to measurable phenotypes. Such limitations become more pronounced when evaluating methane-mitigating feeding strategies, where the goal is to determine how nutritional interventions influence animal performance. The objective of this study was to develop a network-based multi-omics workflow that links transcriptomic variation to phenotypic traits.

Methods & workflow

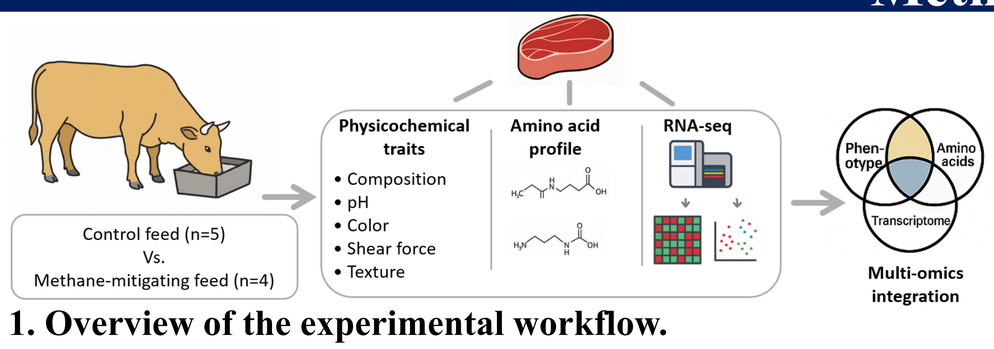

Figure 1. Overview of the experimental workflow.
Hanwoo cattle fed methane-mitigating vs. control feed were evaluated for meat physicochemical traits, amino acid profiles, and RNA-seq. Multi-omics integration was performed to identify feed-responsive molecular signatures.

Table 1. Workflow summary of RNA-seq processing, phenotype profiling, and multi-omics trait–gene integration.

Category	Purpose / Analysis	Tools / Methods	Output
RNA-seq (DEG analysis)	Raw read QC, trimming, alignment to reference genome	FastQC, Trimmomatic, HISAT2/STAR	Clean FASTQ / BAM alignment
	Read quantification	featureCounts (Subread)	Gene count matrix
	Differential gene expression analysis (MM vs Control)	edgeR (GLM QL F-test)	UMAP, Top-genes Heatmap, DEG list
Phenotype / Amino acid profiling	Physicochemical traits (composition, pH, color, shear force, texture) and a mino acid profiling	Lab measurements (AOAC), amino ac id profiling (GC-FID)	Trait dataset
	Identify key traits	Random Forest, MDA/MDG ranking	Radial plot (trait importance)
Multi-omics gene–trait integration	Correlation of transcriptome with phen otype/amino acids	Spearman correlation + BH-FDR	Trait–gene heatmap / bubble plot
	Construct interaction network	R; igraph	Trait–gene network (Figure 5)

Results

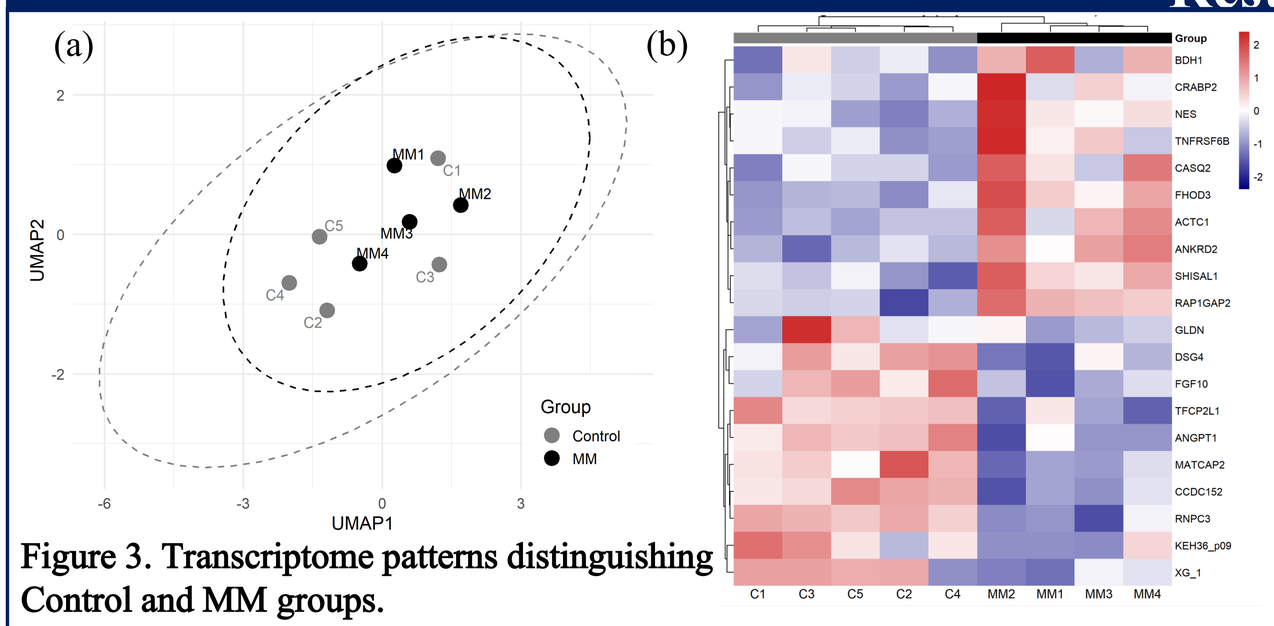

Figure 3. Transcriptome patterns distinguishing Control and MM groups.
(a) UMAP of logCPM (top-variable genes) shows partial separation between groups.
(b) Heatmap of top differentially expressed genes (DEGs) highlights distinct expression signatures between Control and MM.

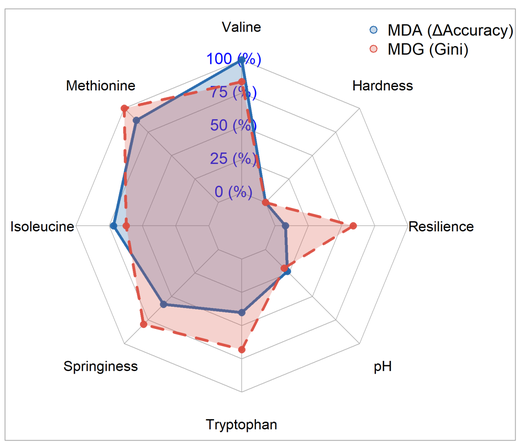

Figure 4. Random Forest trait importance.
Radar plot showing the top traits contributing to group discrimination (MM vs. Control), based on Mean Decrease Accuracy (MDA) and Gini index (MDG).

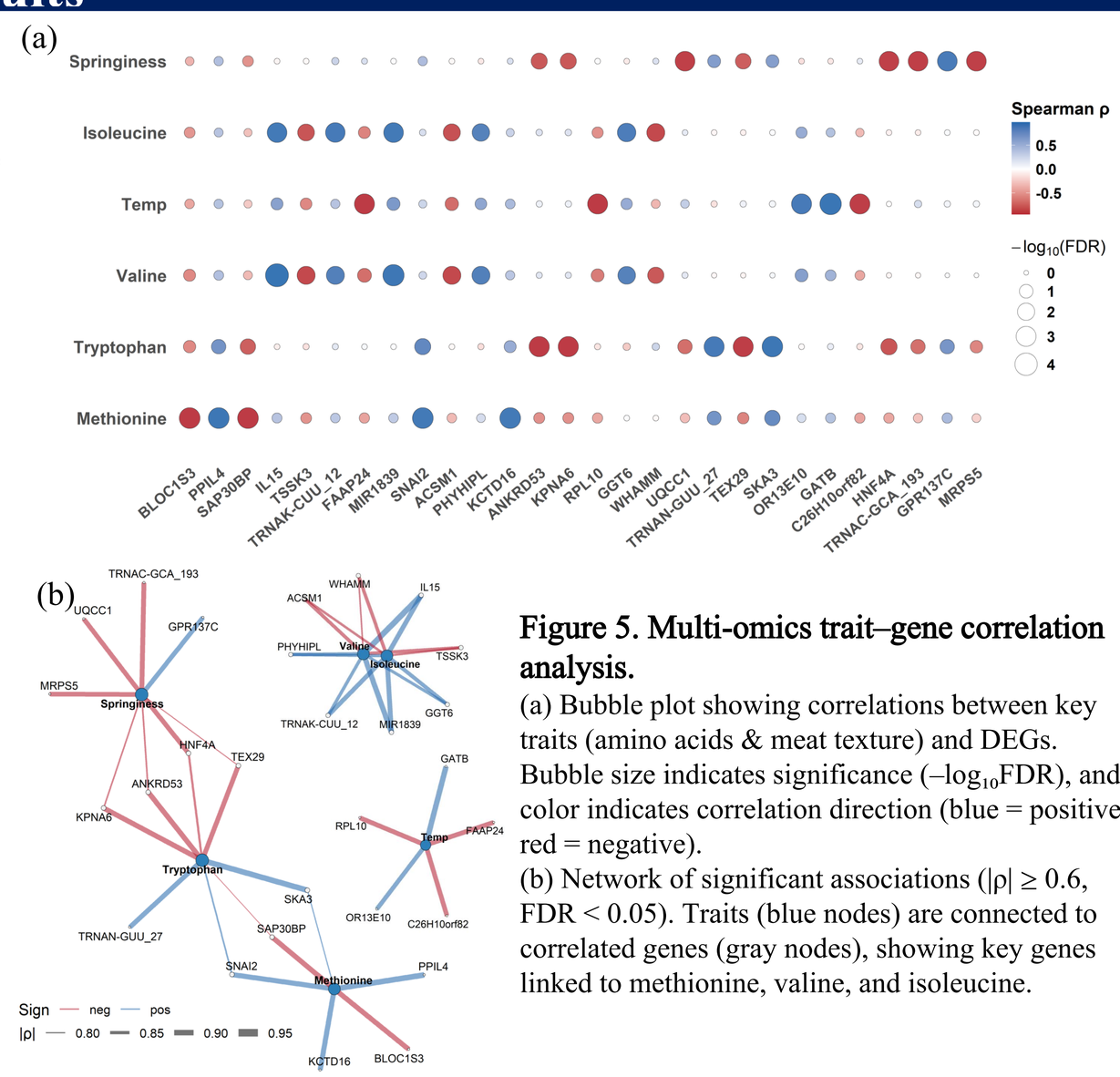

Figure 5. Multi-omics trait–gene correlation analysis.
(a) Bubble plot showing correlations between key traits (amino acids & meat texture) and DEGs. Bubble size indicates significance ($-\log_{10}$FDR), and color indicates correlation direction (blue = positive, red = negative).
(b) Network of significant associations ($|\rho| \geq 0.6$, FDR < 0.05). Traits (blue nodes) are connected to correlated genes (gray nodes), showing key genes linked to methionine, valine, and isoleucine.

Conclusion

- **Transcriptomic variation alone showed limited separation between feeding groups (UMAP).**
- **Network-level integration of gene expression with phenotypic traits (machine learning → correlation → network) revealed biologically meaningful structure.**
- **Gene communities were associated with amino-acid metabolism and muscle-quality traits (e.g., springiness, valine, methionine).**
- **These associations were not detected at the DEG list level but emerged through trait-guided prioritization.**
- **The multi-omics workflow supported biological interpretation rather than simply reporting statistically significant genes**

MM feeding reshapes gene–trait networks linked to muscle amino-acid metabolism and quality.

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